

C720

Operations and Supply Chain Management

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A regional distribution center, which has used the same ERP system for six years, forecasts weekly demand for bottled water using exponential smoothing with $\alpha = 0.2$. Last week's forecast was 500 cases and actual demand was 540 cases. The planner notes a heat wave is expected next month. What is this week's forecast?

- A. 532 cases
- B. 508 cases
- C. Above 540 cases, to get ahead of the heat wave.
- D. 540 cases

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

2. [TRANSFER — reversed stem / different variable]

A planner's previous forecast was 200 units, actual demand came in at 250, and the new exponentially smoothed forecast is 215. What smoothing constant α was used?

- A. 0.3
- B. 0.86
- C. 0.06
- D. 0.7

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

3.

A packaging plant that opened in 2009 has a design capacity of 1,000 cartons a day. After planned maintenance and changeovers its effective capacity is 800 a day, and yesterday it produced 720. The union contract was renewed last spring. What was the plant's efficiency?

- A. 90%
- B. 111%
- C. 72%
- D. 80%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

4. [TRANSFER — reversed stem / different variable]

A bottling line has a design capacity of 1,200 units per hour and ran at 75% utilization. What was its actual output per hour?

- A.** 300 units
- B.** 1,600 units
- C.** 900 units
- D.** 1,200 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A furniture shop builds tables in three steps, each staffed by one worker: cutting takes 12 minutes, assembly 20 minutes, and finishing 15 minutes per table. Cutting uses a new \$80,000 saw. The owner hopes to double sales next year. Running steadily, how many tables per hour can the shop produce?

- A.** It depends on how quickly the new saw pays for itself.
- B.** 3 tables
- C.** About 1.3 tables
- D.** 5 tables

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An urgent care clinic with 8 exam rooms and a recently remodeled lobby sees an average of 30 patients per hour. Each patient spends an average of 0.5 hours from check-in to discharge. On average, how many patients are in the clinic at one time?

- A.** 60 patients
- B.** 30 patients
- C.** 8 patients, one per exam room
- D.** 15 patients

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A warehouse holds an average of 4,000 units and ships 500 units per day. On average, how many days does a unit spend in the warehouse?

- A.** 8 days
- B.** Always 30 days, the standard inventory cycle
- C.** 0.125 days
- D.** 4,000 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A hospital supply room uses 3,600 boxes of gloves a year. Each order costs \$50 to place, and holding one box for a year costs \$4. The supplier, based overseas, charges \$30 per box. What is the economic order quantity?

- A.** 24 boxes
- B.** 110 boxes
- C.** 300 boxes
- D.** 212 boxes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A bike shop sells 40 inner tubes a day. Its supplier, which it orders from through a web portal, delivers 6 days after an order. The shop keeps 60 tubes of safety stock and usually orders 500 at a time. At what inventory level should it reorder?

- A.** 240 tubes
- B.** 106 tubes
- C.** 560 tubes
- D.** 300 tubes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A store's reorder point is 520 units. Daily demand is 30 units and lead time is 15 days. How much safety stock does the reorder point include?

- A.** 970 units
- B.** 450 units
- C.** 70 units
- D.** 520 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Weekly sales of a snack at a grocery chain's registers vary about 5%. The chain's orders to its distributor swing about 20%, and the distributor's orders to the manufacturer swing about 40%. The manufacturer switched packaging suppliers last year. Which action most directly reduces the swings?

- A.** Raise safety stock at every tier.
- B.** Share point-of-sale data with every tier and move to smaller, more frequent orders.
- C.** Treat shopper demand as the source and forecast it more tightly.
- D.** Switch back to the old packaging supplier.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A bottling line that fills 5,000 bottles a shift tracks fill weight on an x-bar chart with a center line of 10.0 oz and control limits of 9.4 and 10.6 oz. The last 20 sample means are all inside the limits with no runs or trends. A new supervisor sees a sample of 10.4 oz and wants to adjust the filler. What is the best response?

- A.** Conclude the line meets customer specifications.
- B.** Adjust the filler down 0.4 oz to re-center it.
- C.** Any sample above the center line means the process is drifting.
- D.** Leave the filler alone; the process is in statistical control.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A regional distribution center, ~~which has used the same ERP system for six years,~~ forecasts weekly demand for bottled water using exponential smoothing with $\alpha = 0.2$. Last week's forecast was 500 cases and actual demand was 540 cases. ~~The planner notes a heat wave is expected next month.~~ What is this week's forecast?

Target word: next-period forecast **Decides:** $500 + 0.2 \times (540 - 500)$.

A	Inverted operation	Puts 0.8 on the actual and 0.2 on the old forecast, reversing the weights.
B	Correct	$500 + 0.2 \times 40 = 508$.
C	Camouflage	The heat wave is next month; the model uses only last period's data.
D	Wrong stem number	Last week's actual, not the smoothed forecast.

2. Answer A Concept 1 · transfer

A planner's previous forecast was 200 units, actual demand came in at 250, and the new exponentially smoothed forecast is 215. What smoothing constant α was used?

Target word: solve for α **Decides:** $215 - 200 = \alpha \times (250 - 200)$.

A	Correct	$15 / 50 = 0.3$.
B	Adjacent category	$215 / 250$ is a forecast-to-actual ratio, not the smoothing constant.
C	Dropped element	$15 / 250$ drops the old forecast from the error term.
D	Inverted operation	That is $1 - \alpha$, the weight on the old forecast.

3. Answer A Concept 2

A packaging plant ~~that opened in 2009~~ has a design capacity of 1,000 cartons a day. After planned maintenance and changeovers its effective capacity is 800 a day, and yesterday it produced 720. ~~The union contract was renewed last spring.~~ What was the plant's efficiency?

Target word: efficiency **Decides:** Efficiency uses effective capacity as the base.

A	Correct	$720 / 800 = 90\%$.
B	Inverted operation	$800 / 720$ flips the ratio.
C	Confusion set	$720 / 1,000$ is utilization, not efficiency.
D	Wrong stem number	$800 / 1,000$ compares the two capacities and ignores output.

4. Answer C Concept 2 · transfer

A bottling line has a design capacity of 1,200 units per hour and ran at 75% utilization. What was its actual output per hour?

Target word: solve for actual output **Decides:** Actual = utilization $\times$ design capacity.

A	Adjacent category	That is the unused (idle) capacity.
B	Inverted operation	$1,200 / 0.75$ divides instead of multiplies.
C	Correct	$0.75 \times 1,200 = 900$.
D	Wrong stem number	Design capacity, not what was produced.

5. Answer B Concept 3

A furniture shop builds tables in three steps, each staffed by one worker: cutting takes 12 minutes, assembly 20 minutes, and finishing 15 minutes per table. ~~Cutting uses a new \$80,000 saw. The owner hopes to double sales next year.~~ Running steadily, how many tables per hour can the shop produce?

Target word: hourly output **Decides:** Assembly at 20 minutes is the bottleneck.

A	Camouflage	The saw's cost does not change step times.
B	Correct	$60 / 20 = 3$; no step can outrun the slowest one.
C	Adjacent category	$60 / 47$ treats total flow time as the cycle time, as if only one table were in process at once.
D	Inverted operation	Uses the fastest step, cutting, instead of the slowest.

6. Answer D Concept 4

An urgent care clinic ~~with 8 exam rooms and a recently remodeled lobby~~ sees an average of 30 patients per hour. Each patient spends an average of 0.5 hours from check-in to discharge. On average, how many patients are in the clinic at one time?

Target word: average number in system **Decides:** $L = \lambda \times W = 30 \times 0.5$.

A	Inverted operation	$30 / 0.5$ divides instead of multiplies.
B	Dropped element	Uses the arrival rate and ignores time in the system.
C	Camouflage	Room count is a capacity detail, not the average number present.
D	Correct	$30 \text{ per hour} \times 0.5 \text{ hour} = 15$.

7. Answer A Concept 4 • transfer

A warehouse holds an average of 4,000 units and ships 500 units per day. On average, how many days does a unit spend in the warehouse?

Target word: solve for time in system **Decides:** $W = L / \lambda = 4,000 / 500$.

A	Correct	$4,000 / 500 = 8$ days.
B	Absolute language	There is no fixed standard; time follows from inventory and throughput.
C	Inverted operation	$500 / 4,000$ flips the ratio.
D	Wrong stem number	Average inventory, not time.

8. Answer C Concept 5

A hospital supply room uses 3,600 boxes of gloves a year. Each order costs \$50 to place, and holding one box for a year costs \$4. ~~The supplier, based overseas, charges \$30 per box.~~ What is the economic order quantity?

Target word: EOQ **Decides:** $\sqrt{(2 \times 3,600 \times 50 / 4)}$.

A	Inverted operation	$\sqrt{(2 \times 3,600 \times 4 / 50)}$ swaps ordering and holding cost.
B	Camouflage	Uses the \$30 price as the holding cost; holding cost is already given as \$4.
C	Correct	$2 \times 3,600 \times 50 = 360,000$; $/ 4 = 90,000$; $\sqrt{} = 300$.
D	Dropped element	$\sqrt{(3,600 \times 50 / 4)}$ leaves out the 2.

9. Answer D Concept 6

A bike shop sells 40 inner tubes a day. Its supplier, ~~which it orders from through a web portal,~~ delivers 6 days after an order. The shop keeps 60 tubes of safety stock ~~and usually orders 500 at a time.~~ At what inventory level should it reorder?

Target word: reorder point **Decides:** $40 \times 6 + 60$.

A	Dropped element	Leaves out safety stock.
B	Inverted operation	$40 + 6 + 60$ adds demand and lead time instead of multiplying.
C	Camouflage	Adds the order quantity, which does not decide when to reorder.
D	Correct	Lead-time demand $240 +$ safety stock $60 = 300$.

10. Answer C Concept 6 · transfer

A store's reorder point is 520 units. Daily demand is 30 units and lead time is 15 days. How much safety stock does the reorder point include?

Target word: solve for safety stock **Decides:** $ROP - \text{lead-time demand} = 520 - 450$.

A	Inverted operation	Adds lead-time demand to the reorder point instead of subtracting.
B	Adjacent category	That is lead-time demand, not the buffer.
C	Correct	$30 \times 15 = 450$; $520 - 450 = 70$.
D	Wrong stem number	The reorder point itself.

11. Answer B Concept 7

Weekly sales of a snack at a grocery chain's registers vary about 5%. The chain's orders to its distributor swing about 20%, and the distributor's orders to the manufacturer swing about 40%. ~~The manufacturer switched packaging suppliers last year.~~ Which action most directly reduces the swings?

Target word: fix for amplified upstream variability **Decides:** Variability grows tier by tier from a stable end demand: the bullwhip effect.

A	Adjacent category	Buffers the symptom at high cost and does not stop the amplification.
B	Correct	Shared demand data and less batching attack the causes of the bullwhip.
C	Confusion set	End demand is the most stable figure; the swings are created upstream.
D	Camouflage	Packaging has nothing to do with order amplification.

12. Answer D Concept 8

A bottling line ~~that fills 5,000 bottles a shift~~ tracks fill weight on an x-bar chart with a center line of 10.0 oz and control limits of 9.4 and 10.6 oz. The last 20 sample means are all inside the limits with no runs or trends. A ~~new supervisor~~ sees a sample of 10.4 oz and wants to adjust the filler. What is the best response?

Target word: act or leave the process alone **Decides:** All points inside limits with no pattern: common-cause variation.

A	Adjacent category	In control is not the same as capable; specs are a separate comparison.
B	Confusion set	Treats a common-cause point as an assignable cause.
C	Absolute language	About half of in-control points fall above center by chance.
D	Correct	Adjusting for common-cause noise is tampering and increases variation.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	3	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Exponential smoothing	New forecast = old forecast + $\alpha \times (\text{actual} - \text{old forecast})$; α weights the most recent error.	Putting α on the old forecast instead of on the newest actual.	A smoothing constant, a prior forecast, and a prior actual in one stem.
2	Utilization vs. efficiency	Utilization = actual output / design capacity; efficiency = actual output / effective capacity.	Using design capacity as the base for efficiency.	Both a design and an effective capacity figure are given.
3	The bottleneck sets throughput	A process can produce no faster than its slowest step.	Using the fastest step or the average step time.	Several step times in series, with a question about hourly or daily output.
4	Little's Law	Average units in the system = arrival (throughput) rate $\times$ average time in the system.	Dividing when you should multiply, or mixing time units.	A rate per time unit plus a wait or flow time.
5	Economic order quantity	$EOQ = \sqrt{(2DS / H)}$; it balances ordering cost against holding cost.	Dropping the 2, swapping S and H, or plugging in unit price.	Annual demand, cost per order, and holding cost per unit per year.
6	Reorder point	Reorder point = daily demand $\times$ lead time (days) + safety stock.	Leaving out safety stock or confusing it with the order quantity.	Lead time in days next to daily usage.
7	Bullwhip effect	Small swings in end-customer demand grow larger at each upstream tier; the fix is shared demand data and smaller, steadier orders.	Blaming end customers or piling on more safety stock everywhere.	Order variability that grows tier by tier moving upstream.
8	Common vs. assignable cause variation	Points inside control limits with no pattern are common-cause noise; adjusting for them (tampering) adds variation.	Reacting to every point that is off the center line.	A control chart with all points inside limits and a manager who wants to adjust.

Course scope (public description): Operations and Supply Chain Management provides a streamlined introduction to how organizations efficiently produce goods and services, determine supply chain management strategies, and measure performance. Emphasis is placed on integrative topics essential for managers in all disciplines, such as supply chain management, product development, and capacity planning. This course guides students in analyzing processes, managing quality for both services and products, and measuring performance while creating value along the supply chain in a global environment. Topics include forecasting, product and service design, process design and location analysis, capacity planning, management of quality and quality control, inventory management, scheduling, supply chain management, and performance measurement.